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Examining how agricultural lands in the Wildland-Urban Interface can affect community resilience

Introduction

Wildfires play a very important role in California ecology, increasingly more so with the escalation of human-induced climate change and thus the increase in landscapes having conditions suitable for fire. Fire services are pertinent to the health of our ecosystems, but they pose a large threat to structures, especially at wildland-urban interfaces (WUIs) where the built and natural environments converge. Included in WUIs are agricultural lands, which make up a significant portion of the WUIs in California as about 40% of land use in the state is for agriculture (PPIC, 2024). Though farmland itself is not largely burned, crops, livestock, and agricultural workers face threats from other fire impacts. Crops can be damaged from smoke and heat, and poor air quality conditions pose a serious health risk for agricultural workers and farm owners. The effective management of agricultural lands is necessary to reduce risks as much as possible. We will examine the relationship between wildfire and agriculture, the historical context of agricultural WUI fires, and how producers can move forward in increasing resilience for themselves and their communities.

Historical Fire Management

California's Central Valley is home to around 75% of the state's irrigated land (USGS). Since this huge conversion of land usage over the past 200 years, fires are not seen as an ecosystem function for the area, apart from the burning of excess crop residue and land-clearing agricultural uses. Despite the seemingly unfamiliar role that wildfires play in the Central Valley, prior to European settlement, fires were commonly used as a management tool. And as lightning strikes are very uncommon in this region of California, fire presence in general has been primarily due to Indigenous activity. The Central Valley pre-colonization and pre-land-conversion consisted of grasslands with a large presence of wetlands, like riparian woodlands, freshwater marshes, and vernal pools (Barry et al., 2006). These wetland areas were managed by Indigenous peoples, which included the use of fire for clearing away flora. Fires were also used in order to bolster herpetofauna and small mammal populations during the beginning of the wet season (October to January) in the Central Valley. Populations like deer mice and western fence lizards typically thrive in open areas, so utilizing fires to clear brush allowed for their habitat to foster population growth. A 2009 study done in remaining riparian ecosystems in the northern Central Valley tested this relationship between fire usage and deer mice and fence lizard populations (Hankins, 2013). Local Indigenous practitioners were consulted throughout the study for knowledge about fire usage. First, ladder fuels were removed through pruning and coppicing before conducting the burns in order to replicate the understory of pre-colonization. Fires were then set using drip torches, and were ignited at random intervals in an attempt to mimic Indigenous practices from the area. Post-burn deer mice and fence lizard populations were compared to control sites, and it was concluded that land management via fire does increase these populations. This conclusion gives even greater purpose to Indigenous groups in California's Central Valley as the impacts are seen beyond the typical clearing of flora and opening of space for megafauna.

In the grasslands of the Central Valley (and in the Coast Ranges), fire was used as a tool by Indigenous populations. These prairies were burned annually in order to halt their development into forests; doing so allowed for seeds to be harvested and for large game to graze there (Hankins, 2013). In these riparian and grassland ecosystems of the Central Valley, fire was used as a tool for managing resources. Now, due to land conversion and a massive scale of agriculture, fires as ecosystem management tools are no longer seen where they once were. The remaining California grasslands that are not undergoing land-use-change are facing a reduction in acreage and overall quality due to the encroachment of woody and nonnative species, indicating that a reintroduction of burning practices may be beneficial (Barry et al., 2006).

How Agriculture is Affected by Fire

Other than direct burning from wildfires, producers face crop damage from smoke and ash, health impacts, business disruption, infrastructure damage, and the degradation of natural resources. A survey conducted during the summer of 2023 asked producers about their experiences with fire (Pinzón et al., 2025). Of the respondents, 90% reported being affected by smoke and/or ash, 45% reported being affected by fire near their operation, and roughly half reported being affected by power outages and evacuation orders. Smoke and ash can be a serious threat to human health, and as an industry based around agricultural workers being outside, their safety must be of top consideration. Around 35% of the crop losses that were reported are from the inability to work, harvest, and/or irrigate due to air conditions. The air quality extends to livestock as well, with 62% of livestock producers reported negative impacts on their animals' health and production. Even greater crop losses were suffered due to ash damage, making up 46% of total losses. Ash damage manifests through the blockage of sunlight to leaves, through physical damage as the ash can build up and become heavy, and through impacting the soil chemistry. These impacts can lead to the decreased growth of crops, decreased overall yield, and an actual alteration to the crop flavor or appearance, most often rendering it not fit to sell. The amount of crops affected by direct fire exposure was reported to be 17%. These crops were either burned, singed, or dried out from the heat, which would all render the crop unmarketable. Some of the most striking impacts is that crop loss was, in the most affected cases, sustained for future seasons. Producers that suffered from fire on their land reported that 20% of the crop losses persisted past the season of the fire. Some producers reported experiencing benefits from the fire, with over a quarter observing an increase in flowering and pollinator activity, but overall, fire near or on an agricultural operation can lead to serious crop losses and health risks.

The Role of Agricultural Lands and Producers in Fighting Fires

In terms of protection during a wildfire, agricultural lands are often low on the list for firefighters compared to saving lives and protecting other properties, such as residential homes or businesses. However, agricultural lands, which can range from irrigated croplands to livestock-grazed grasslands, can serve as important buffers against wildfires due to their position in or near the wildland-urban interface, their constant management prior to a fire, and resources there that can assist firefighters on the frontline in battling approaching fires.

When surveyed about the effects of wildfires and their response to them over the course of 7 years from 2017 to 2023, California agricultural producers reported their frontline response and assistance to emergency responders in many different ways. Out of 505 producers surveyed, 303 (77%) provided assistance to emergency responders as well as community members in need in the vicinity of wildfires. This percentage grew to 91% for the 128 producers in the survey who experienced fire directly on their farm (Pinzón et al., 2025). Some of the assistance producers provided, regardless of whether they experienced fire directly or indirectly, included creating firebreaks and combating the flames directly, sharing specific, local knowledge about the terrain with firefighters, providing water and supplies, and evacuating people and livestock. Additionally, defense of their own property was a crucial point for many farmers, with 65% of them reportedly staying on their lands at least once in order to defend against encroaching fires. Given the importance and value of their land, infrastructure, equipment, and livestock for their livelihood, this response is fairly unsurprising. However, this fact can lead to the expansion of future collaboration between farmers and firefighters that can benefit both parties, while leading to more robust community protection. In the survey, many producers also expressed the desire for more available resources in order to defend their land more effectively. Among them included better water systems, water storage, generators for when power is inadvertently shut off, crew training, and better overall community preparedness (Pinzón et al., 2025). Another desired resource was the expansion of emergency passes that would allow farmers to return to their farms amid evacuation orders, which we will discuss later in more detail. Investment in these resources would vastly increase the ability for producers to support emergency crews as well as reduce the risk of them remaining on the frontline of the wildfire. Due to the extreme danger that comes from staying so close to the fire, as well as not being firefighters themselves, producers would not be expected to act in the same capacity as emergency responders. Yet, this type of system can provide a framework for community-based defense against wildfire that is becoming increasingly necessary as wildfires in California have increased in severity and frequency over the past decades. In times of crisis where emergency responders simply do not have the resources or manpower available to cover every community under threat, these kinds of community-based plans can mean the difference between being able to save entire communities and even the lives of its inhabitants.

Another aspect of agricultural lands that makes it both a good buffer against wildfire and a good staging ground for firefighters is the fuel reduction management that already happens on most farmland. Fuel reduction in this case refers to the removal of fine fuels in and around infrastructure and other vegetation that might easily ignite and start a spot fire. For the most part, massive loss of human infrastructure in the wildland-urban interface is due to home ignitability from flammable roofing materials and burnable vegetation adjacent to homes (Cohen, 1999). Given that agricultural lands are less densely populated with houses that could be made of flammable materials, it makes it much more difficult for a high intensity fire to start there. This, combined with the already mentioned fuel reduction that most farms perform in order to clear space for their crops, often turns farms and ranches into safe spots amid encroaching flames. This kind of practice is referred to as defensible space, which is described as the buffer between a structure and the surrounding area [and] can act as a barrier to slow or halt the progress of fire (Cal Fire). Establishment of defensible space around structures has become an incredibly important tool for people living in the wildland urban interface in recent years, being a relatively feasible task for most homeowners and being relatively effective against home ignition. Unsurprisingly, in the survey mentioned above, 72% of producers reported taking action to create defensible space around their property. Methods to do so include fuel management through forest thinning, grazing, and even small-scale prescribed burns (Pinzón et al., 2025). It is worth noting that fuel reduction management, while it has a lot of potential and already-proven benefits, requires that it is done on a landscape-level; a local-level extent for fuel reduction is not very effective in changing fire behavior. Generally speaking, fuel treated areas do not inhibit fire spread or ignition significantly if treatment areas were small or narrow, the treatment did not reduce the fuel load enough, or implementation was not complete (Urza et al., 2023). This makes it important that farmers implement these kinds of fuel treatments across the entire landscape, not limiting it to pockets of farmland in a scattered mosaic. As stated previously, allocating more resources in order to coordinate all of the producers in the region to create a shared community-based defense plan will be crucial in order to protect people and their property in this region.

Roadmap Forward

From this, one of the most promising methods for agricultural land resilience is the previously mentioned Ag pass system due to its achievable nature in the short-term but also its potential in the long-term. As established, farmers are often responsible for protecting their own lands due to the lower priority of saving their sparsely populated orchards and grasslands by emergency responders. Yet, investing more resources into protecting these lands can net both producers and firefighters immense benefits and relief in the face of crises. In this case, resources does not refer solely to financial investment of better water and power infrastructure, but also the expansion of community-based plans that can lead to better organization between the many agencies and groups that respond to a wildfire. An Ag pass is a standardized way for emergency personnel to identify farm and ranch owners in the locale who may need to return to their lands amid evacuation orders to protect it as well as offer support to first responders (County of Santa Barbara). This kind of system already exists for media reporters gaining access to restricted areas in order to document disasters under California Penal Code 409.5, but an Ag pass system is specifically for agricultural workers. The point of an Ag pass program is to have this type of framework already in place before a disaster occurs in order to better coordinate response efforts and avoid confusion between parties.

Another strong point about an Ag pass program is that it can incentivize training for responding to wildfires among agricultural workers, which can strengthen community resilience and preparation against it. This program has the potential to formalize on-going education and training for producers as a requirement for renewing their Ag passes, teaching health and safety practices, shelter-in-place protocols, and many other disaster practices (Shapero et al., 2020). An advantage of having it as an on-going program is that it can teach communities about up-to-date practices as advised by agencies such as Cal Fire, preventing the spread of outdated information. That being said, this ongoing training, while extremely important, could become a barrier to participation in this program as it would mean producers would have to be committed to some degree to this program.

There have already been several use cases of the Ag pass program in the vicinity of Santa Barbara, which revealed both its effectiveness and some of its shortcomings. One example of this can be found in neighboring Ventura County. Following the 1985 Wheeler Fire, which burned 120,000 acres in Ventura County (Cal Fire) and damaged many orchards and ranches around Ojai, discussion emerged about a system to allow producers to return to their lands amid a fire in order to protect them. This plan was finally implemented in a sort of prototype Ag pass system in 2000 with the Ventura County Agricultural Worker ID Program and expanded out into the Ventura County Ag Pass Program by 2017 (Shapero et al., 2020). By this time, despite the expansion of the program and the establishment of an official vetting process, it still lacked coordination with many different agencies, such as California Highway Patrol and the Ventura County Fire Department, who often lacked knowledge about this program. This shortcoming became exceedingly apparent during the 2017 Thomas Fire, which burned an approximate 280,000 acres across Ventura and Santa Barbara counties (Cal Fire). Lack of awareness from emergency responders, combined with the fact that many fire crews came from out of county to assist with fire suppression due to the fire's immense scale, meant that usage of the Ag pass was very inconsistent. Many producers reported agricultural losses after the fire due to being turned away at road closures, meaning they could not properly irrigate their lands and evacuate their livestock. On the other hand, producers who were able to use the Ag pass to access their lands reported success in being able to effectively defend their lands using irrigation and offer assistance to firefighters by locating and identifying key defensive points on the landscape. This case shows the benefits of implementing Ag pass programs as well as some of the challenges that come from using such a system.

An informational hearing held by the Assembly Committee on Agriculture in November of 2020 brought up many of the impacts and concerns held by California producers in regards to the previous five years' wildfires. After discussing the impacts faced, a list of future actions proposed by the panelists was consolidated and policy considerations were drafted. Many individuals mentioned the increasing rates of fire insurance, the lack of availability of protection equipment and training, a proposal for the increase of grazing, and the expansion of prescribed burns. Policy action ideas discussed included greater clarity of the meaning of "farm risk" and expanding insurance to be able to include farm buildings (CA State Assembly, 2020). For protective equipment, AB 73 was introduced following the hearing, and this act would allow for the creation of a backup stock of PPE and the requirement for agricultural workers to undergo wildfire smoke training. The expansion and incentivization of grazing on public lands was also desired as it can aid wildfire prevention/mitigation. Lastly, proposed policy action for prescribed burning included adopting a liability standard for gross negligence for prescribed fires as liability and insurability concerns are some of the largest barriers for controlled burning. Additionally, it was suggested that the amount of "no-burn" decisions that are made for the concern of health impacts should be decreased (as smaller-scale, prescribed burns would overall create less harmful conditions than an undesired, over-fueled wildfire) and that certain prescribed burns should be exempt from CEQA requirements. These requests, in conjunction with Ag passes, should be considered for future actions protecting California producers and their agricultural lands.

Conclusion

Agricultural land management and defense in the wildland-urban interface can have profound effects on creating a buffer against wildfires in the area, leading to better community-based resilience and preparedness in the face of ever-growing wildfire concern in California. Agricultural lands are not only of great economic importance in California, making them important to protect, but also represent a unique opportunity to defend the entire region and its inhabitants from fire through the nature of its landscape, the resources it can provide to emergency responders, and the management it undergoes on a daily basis. Many of the methods for community resilience described above have already seen some success across the state and through the years, but its effectiveness is hampered by limited resources, participation, and communication. By raising awareness for some of these community-based plans for wildfire defense, communities along the wildland-urban interface can better prepare for disaster without relying completely on emergency services which may be overwhelmed.

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